Grade 7 Mathematics

Strand 1: Number

Sub-strand 3: Fractions, Decimals and Percentages

Lesson 1 — Fractions, Decimals and Percentages in Full

Codes: B7.1.3.1.1–1.2, B7.1.3.2.1–2.2, B7.1.3.3.1–3.4

Teaching order inside this block: benchmark equivalences → compare & order → add/subtract → multiply → fraction of a quantity → divide (and dividing a quantity).

1. What this lesson is about

A **fraction** is a way to write part of a whole: `(3)/(8)` means 3 parts, each one eighth.

This lesson covers three faces of the same idea:

Face	Example
Fraction	`(1)/(2)`
Decimal	0.5
Percentage	50%

Words to unpack first:

Word	Meaning
Numerator	top number (how many parts we have)
Denominator	bottom number (how many equal parts the whole was cut into)
Percentage	"out of 100"
Benchmark fractions	halves, thirds, quarters, fifths, eighths, tenths
Common denominator	the same bottom number
LCM	lowest common multiple — the smallest shared multiple of the denominators

2. Benchmark equivalences (B7.1.3.1.1)

Learn these cold — they make everything else faster:

Fraction	Decimal	Percentage
`(1)/(2)`	0.5	50%
`(1)/(4)`	0.25	25%
`(3)/(4)`	0.75	75%
`(1)/(5)`	0.2	20%
`(1)/(8)`	0.125	12.5%
`(1)/(3)`	0.333…	33■%
`(1)/(10)`	0.1	10%

Converting:

- Fraction → decimal: **divide top by bottom** ((3)/(8) = 3 ÷ 8 = 0.375).
- Decimal → percentage: **multiply by 100** (0.375 → 37.5%).
- Percentage → fraction: **divide by 100**, then simplify (37.5% = 37.5/100 = 3/8).

3. Compare and order (B7.1.3.1.2)

Same denominator: bigger numerator wins ((5)/(8) > (3)/(8)).

Different denominators: use a common denominator (LCM).

Example: order (3)/(8), (1)/(2), (5)/(16). LCM of 8, 2, 16 is 16.

(3)/(8) = (6)/(16), (1)/(2) = (8)/(16), (5)/(16) = (5)/(16)

Order: (5)/(16) < (3)/(8) < (1)/(2).

Cross-multiply shortcut: to compare (a)/(b) and (c)/(d), compare ad and cb.

4. Add and subtract fractions (B7.1.3.2)

You **must** have the same denominator.

(1)/(3) + (1)/(4) = (4)/(12) + (3)/(12) = (7)/(12)

Mixed numbers: convert to improper fractions first.

2(1)/(2) - (1)/(3) = (5)/(2) - (1)/(3) = (15-2)/(6) = (13)/(6) = 2(1)/(6)

Word problem (B7.1.3.2.2): Amina had 2(1)/(2) litres and drank (1)/(3) litres. Left? → 2(1)/(6) litres.

5. Multiply fractions (B7.1.3.3.1–3.2)

Top × top, bottom × bottom (cancel first if you can):

(2)/(3) × (1)/(4) = (2)/(12) = (1)/(6)

Fraction of a quantity:

(3)/(8) of 480 = (3)/(8) × 480 = 180

6. Divide fractions (B7.1.3.3.3–3.4)

Keep, Change, Flip (multiply by the reciprocal):

(2)/(5) ÷ (14)/(15) = (2)/(5) × (15)/(14) = (30)/(70) = (3)/(7)

Dividing a quantity: How many (1)/(4)-litre cups in 3 litres? 3 ÷ (1)/(4) = 3 × 4 = 12 cups

7. Twenty fully worked examples

1. (1)/(2) = 0.5 = 50%.
2. (3)/(8) = 0.375 = 37.5%.
3. 0.25 = (1)/(4) = 25%.
4. 12.5% = 0.125 = (1)/(8).
5. Order (3)/(8), (1)/(2), (5)/(16) → (5)/(16) < (3)/(8) < (1)/(2).
6. (3)/(8) + (1)/(8) = (1)/(2).
7. (1)/(3) + (1)/(4) = (7)/(12).
8. (5)/(6) - (1)/(4) = (7)/(12).
9. 2(1)/(2) - (1)/(3) = 2(1)/(6).

10. (2)/(3) × (1)/(4) = (1)/(6).
11. (3)/(8) of 480 = 180.
12. 35% of GH■ 240 = 84.
13. (2)/(5) ÷ (14)/(15) = (3)/(7).
14. 3 ÷ (1)/(4) = 12 cups.
15. (1)/(2) of (3)/(5) = (3)/(10).
16. Convert 2(1)/(2) to improper → (5)/(2).
17. Convert (11)/(4) to mixed → 2(3)/(4).
18. 45% vs (9)/(20): (9)/(20) = 45% → equal.
19. (1)/(2) + (1)/(3) × (3)/(4) = (1)/(2) + (1)/(4) = (3)/(4) (BODMAS).
20. ≤ft((1)/(2) + (1)/(3)) × (6)/(5) = (5)/(6) × (6)/(5) = 1.

8. Eight exercise sets × 5 questions

Exercise 1 — Equivalences

1. Write (1)/(2) as a decimal and a percentage.
2. Write (3)/(8) as a decimal and a percentage.
3. Write 0.25 as a fraction and a percentage.
4. Write 12.5% as a fraction.
5. **[Review]** 2^3 × 2^4.

Exercise 2 — Compare and order

1. Order (3)/(8), (1)/(2), (5)/(16).
2. Which is larger: (5)/(8) or (3)/(5)?
3. Order (1)/(3), (2)/(5), (1)/(2).
4. Compare 45% and (9)/(20).
5. **[Review]** Round 12.473 to 1 d.p.

Exercise 3 — Add and subtract

1. (3)/(8) + (1)/(8)
2. (1)/(3) + (1)/(4)
3. (5)/(6) - (1)/(4)
4. 2(1)/(2) - (1)/(3) (Amina's water)
5. **[Review]** 124.3 + 0.275 + 74.06 to 1 d.p.

Exercise 4 — Multiply

1. (2)/(3) × (1)/(4)
2. (1)/(2) of (3)/(5)
3. 3(1)/(3) × (3)/(10)
4. (7)/(8) × (4)/(21) (cancel first)
5. **[Review]** Value of 6 in 9 463 285.

Exercise 5 — Fraction of a quantity & percentages

1. (3)/(8) of 480
2. 35% of GH■ 240
3. A shirt marked GH■ 80 is sold at a 15% discount. Find the sale price.
4. (1)/(4) of 1 000
5. **[Review]** Round 84 730 to 2 s.f.

Exercise 6 — Divide fractions

1. `(2)/(5) ÷ (14)/(15)`
2. `3 ÷ (1)/(4)` (how many quarter-litre cups in 3 litres?)
3. `(3)/(4) ÷ ≤ft((1)/(2) - (1)/(4))`
4. `(5)/(6) ÷ (5)/(12)`
5. **[Review]** `5^6 ÷ 5^2`.

Exercise 7 — Convert mixed ↔ improper

1. `2(1)/(2)` → improper
2. `(11)/(4)` → mixed
3. `3(2)/(5)` → improper
4. `(17)/(5)` → mixed
5. **[Review]** Write 3 050 082 in words.

Exercise 8 — Mixed / challenge (BODMAS)

1. `(1)/(2) + (1)/(3) × (3)/(4)`
2. `≤ft((1)/(2) + (1)/(3)) × (6)/(5)`
3. `(2)/(3) × (1)/(4) + (2)/(5) ÷ (14)/(15)`
4. **[Review]** `48 ÷ 6 + 5 × 2 - 3`
5. **[Review]** `(3)/(8) + (1)/(4)`.

9. Full worked solutions

Solutions — Exercise 1

1. `0.5 = 50%`. 2. `0.375 = 37.5%`. 3. `(1)/(4) = 25%`. 4. `(1)/(8)`. 5. **128**.

Solutions — Exercise 2

1. `(5)/(16) < (3)/(8) < (1)/(2)`. 2. `(5)/(8)`. 3. `(1)/(3) < (2)/(5) < (1)/(2)`. 4. **equal**. 5. **12.5**.

Solutions — Exercise 3

1. `(1)/(2)`. 2. `(7)/(12)`. 3. `(7)/(12)`. 4. `2(1)/(6)` litres. 5. **198.6**.

Solutions — Exercise 4

1. `(1)/(6)`. 2. `(3)/(10)`. 3. 1. 4. `(1)/(6)`. 5. **60 000**.

Solutions — Exercise 5

1. **180**. 2. **GH■ 84**. 3. **GH■ 68**. 4. **250**. 5. **85 000**.

Solutions — Exercise 6

1. `(3)/(7)`. 2. **12 cups**. 3. **3**. 4. 2. 5. **625**.

Solutions — Exercise 7

1. `(5)/(2)`. 2. `2(3)/(4)`. 3. `(17)/(5)`. 4. `3(2)/(5)`. 5. **three million, fifty thousand, eighty-two**.

Solutions — Exercise 8

1. `(3)/(4)`. 2. **1**. 3. `(25)/(42)`. 4. **15**. 5. `(5)/(8)`.

10. Cheat sheet

Skill	Rule
Compare fractions	common denominator
Add / subtract	same denominator first
Multiply	top×top, bottom×bottom
Divide	Keep, Change, Flip
Fraction of quantity	fraction × quantity
Fraction → decimal	divide top by bottom
Decimal → %	× 100

NaCCA CCP B7.1.3 — plain English, British notation.